Euihyeon Choi Ph.D. Student

Last Update: Aug. 18, 2026

Daniel Guggenheim School of Aerospace Engineering
Georgia Institute of Technology
Website: euihyeonchoi.github.io
LinkedIn: linkedin.com/in/ehchoi
Email : ehchoi@gatech.edu
Phone : +1-404-860-9436

Education

Georgia Institute of Technology — Atlanta, GA, USA
Ph.D. in Aerospace Engineering, Expected August 2027 — Aug. 2024 – Aug. 2027 (Expected)
- Advisor: Prof. Koki Ho

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Republic of Korea
Master of Science in Aerospace Engineering — Aug. 2022 – Aug. 2024
- Thesis: Optimal planetary surface routing considering synergy value system and heterogeneous agents
- Advisor: Prof. Jaemyung Ahn

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Republic of Korea
Bachelor of Science in Aerospace Engineering — Mar. 2015 – Feb. 2019
Bachelor of Science in Physics (Double Major)

Professional and Research Experience

Georgia Institute of Technology — Atlanta, GA, USA
Graduate Research Assistant, Daniel Guggenheim School of Aerospace Engineering — Aug. 2024 – Present
Graduate Teaching Assistant (Spacecraft Flight Dynamics) — Aug. 2026 – Present

Korea Advanced Institute of Science and Technology — Daejeon, Republic of Korea
Graduate Research Assistant, Department of Aerospace Engineering — Aug. 2022 – Aug. 2024

Agency for Defense Development (ADD) — Daejeon, Republic of Korea
Researcher, Aerospace Technology Research Institute — Jun. 2019 - May. 2022
Research Officer for National Defense, Republic of Korea Air Force

Publications

Journal Articles

[J1] **E. Choi** and J. Ahn, "Optimal Routing for In-Situ Resource Utilization-Based Planetary Surface Explorations," *Journal of Aerospace Information Systems*, 2026 (Accepted)

[J2] **E. Choi** and K. Ho, "Formulation and Analysis for Integrated Spacecraft Routing and Trajectory Design Problem," *Journal of Spacecraft and Rockets*, 2026 (Accepted)

[J3] B. Gwon*, **E. Choi***, J. Lee, and J. Ahn, "Adaptive Planning for Multiple UAVs with In-Flight Refueling," *International Journal of Aeronautical & Space Sciences*, 2026 (Accepted)
**These authors contributed equally.*

[J4] J. Choi, J. Lee, **E. Choi** and J. Ahn, "Sequential Vertiport Location Optimization Considering Uncertain Social Benefit," *Journal of Aerospace Information Systems*, 2026, pp. 1–16

[J5] **E. Choi** and K. Ho, "Orbital Depot Location Optimization for Satellite Constellation Servicing with Low-Thrust Transfers," *Journal of Spacecraft and Rockets*, Vol. 63, No. 1, 2026, pp. 4–13.

[J6] C. Moon, **E. Choi**, and J. Ahn, "Aerial-Ground Collaborative Routing with Uncertain Exploration Value," *International Journal of Aeronautical & Space Sciences*, Vol. 26, 2025, pp. 2817–2841.

[J7] J. Kim, **E. Choi**, J. Ahn, E.S. Suh, J. Kim, and D.G. Lim, "Mathematical Programming-Based Design Structure Matrix Clustering for Modular Architecture Design," *Journal of Mechanical Design*, Vol. 147, No. 10, 2025, 101401.

[J8] **E. Choi** and J. Ahn, "Optimal Multi-agent Grouping and Routing with Cooperative Tasks," *Journal of Aerospace Information Systems*, Vol. 22, No. 3, 2025, pp. 163–176.

[J9] **E. Choi** and W. Chang, "Multi-Agent Task Allocation with Inter-Agent Distance Constraints," *Journal of Aerospace Information Systems*, Vol. 21, No. 2, 2024, pp. 168–177.

[J10] **E. Choi** and J. Ahn, "Optimal Planetary Surface Exploration with Heterogeneous Agents," *Journal of Spacecraft and Rockets*, Vol. 60, No. 4, 2023, pp. 1343–1354.

[J11] **E. Choi** and J. Ahn, "Optimal Planetary Surface Routing Considering Exploration Values and Synergies," *Journal of Aerospace Information Systems*, Vol. 19, No. 6, 2022, pp. 406–420.

Working Papers

[P1] B. Gwon, **E. Choi**, and J. Ahn, "An Integer Linear Programming-Based Framework for Satellite Coverage Optimization Problem Under Operational Constraints," *International Journal of Aeronautical & Space Sciences*, (Major revision)

[P2] **E. Choi**, J. Lee, and J. Ahn, "Multi-Commodity Lunar Campaign Optimization via Leveraging Hybrid Propulsion Systems and Resonance Orbits," *Journal of Spacecraft and Rockets*, (Major revision)

[P3] **E. Choi**, J. Kim, J. Kim, J. Ahn, "Optimal Design Structure Matrix Clustering with Constraint Selections for Conflict Resolution," *Journal of Mechanical Design*, (Under review)

Conference Papers

[C1] **E. Choi** and K. Ho, "In-Space Recycling Logistics Optimization for Circular Space Economy," *2027 AIAA SciTech Forum,* (Submitted).

[C2] T. Noro, **E. Choi**, S. Han, M. Isaji, and K. Ho, "Optimal Scheduling of ADR Missions in the Near-GEO Region Based on a Competing-Risk Failure Model," The 70th Space Sciences and Technology Conference, November 2026.

[C3] T. Noro, K. Ho, **E. Choi**, S. Han, and M. Isaji, "Low-Thrust Trajectory Optimization via Sequential Convex Programming for On-Orbit Servicing in GEO," The Japan Society for Aeronautical and Space Sciences 57th Annual Meeting, April 2026.

[C4] **E. Choi** and K. Ho, "Optimal Routing and Trajectory Planning for Asteroid Mining with Partial In-Situ Resource Utilization," *2026 AIAA SciTech Forum,* January 2026.

[C5] **E. Choi** and J. Ahn, "Optimal Routing Problem with Precedence Constraints for Surface Exploration," *2024 AIAA SciTech Forum,* January 2024.

[C6] J. Choi, J. Lee, **E. Choi** and J. Ahn, "Optimizing UAM Vertiport Location in Jeju Island: Incorporating Noise for Public Utility," *2023 Asia-Pacific International Symposium on Aerospace Technology,* October 2023.

[C7] **E. Choi** and J. Ahn, "Optimal Short-Term Scheduling for Space Telescope in Low-Earth Orbit," *2023 AAS/AIAA Astrodynamics Specialist Conference,* August 2023.

[C8] **E. Choi**, D. Han, and W. Chang, "Decentralized Task Allocation for Multiple UAVs with Iterative Cost Reduction," *KSAS 2023 Spring Conference,* pp. 60–61, April 2023.

[C9] J. Ahn, **E. Choi** and D.U. Lee, "Application of routing problems to space exploration missions," *2023 AIAA SciTech Forum,* January 2023.

[C10] W. Chang and **E. Choi**, "Trade-off Study on Multi-UAV Autonomous Mission Planning Algorithms," *KSAS 2022 Fall Conference,* pp. 1736–1737, 2022.

[C11] **E. Choi** and W. Chang, "Multi-Agent Decentralized Task Allocation Algorithms with Fuel Constraints," *KSAS 2021 Fall Conference*, pp. 432–433, 2021.

Media Articles

[M1] **E. Choi** and K. Ho, "SpaceX aims to use rockets to quickly transport cargo across the Earth and into orbit – moves that would expand the global cargo system," *The Conversation*, 2026.

Patents and Software Disclosures

1. T. Noro, M. Isaji, **E. Choi**, S. Han, and K. Ho, "PosbasedOOSnearGSO.jl," Software Disclosure, Invention ID 2026-248, 2025.
2. T. Noro, M. Isaji, **E. Choi**, S. Han, and K. Ho, "LowthrustSCP.jl," Software Disclosure, Invention ID 2026-247, 2025.
3. D. Han, W. Jang, **E. Choi**, "Method and apparatus for determining the path of an unmanned vehicle using a distributed network," *KR Patent*, 10-2946314, 2026.
4. **E. Choi**, K. Ho, and Y. Shimane, "Continuous Location Routing Problem for Orbital Depot (CLRPOD) v1.0.0," Georgia Tech Software Disclosure, Invention ID 2026-018, 2025.
   – https://github.com/EuihyeonChoi/CLRPOD

Scholarships and Awards

- **Global Korea Scholarship Program for Study Abroad (Ph.D.) by Korean Government** — Fall 2024 – Summer 2026
  Annually $40,000 and Airfare
- **Finalist in 2023 Space Grand Challenge by KAIST, Hanwha Aerospace, and Hanwha Systems** — Nov. 2023
- **KAIST Scholarship** — Fall 2022 – Spring 2024
  Annually $15,000 (Full-Ride)
- **ADD President's Award: The Best Research Proposal** — Dec. 2019
  Subject: Battery Dumping System to Improve Drone's Endurance
- **Degree of Bachelor of Science in Cum Laude** — Feb. 2019
- **Finalist in NASA's RASC-AL Competition** — Cocoa Beach, FL, USA
  Theme: Design Lunar Polar Sample Return Architecture — Aug. 2017 – Jun. 2018
- **National Natural Science and Engineering Scholarship** — Spring 2017 – Fall 2018
  Total $20,000 (Full-Ride)
- **The Boeing Company Scholarship** — Spring 2016 – Fall 2017
  Total $3,500
- **National Scholarship for Undergraduate Study** — Spring 2015 – Fall 2016
  Total $13,500 (Full-Tuition)

Technical Journal or Conference Referee Activities

- Reviewer, *AIAA Journal of Aerospace Information Systems*, 2025, 2026
- Reviewer, *International Journal of Aeronautical and Space Sciences*, 2026
- Reviewer, *Cluster Computing*, 2026
- Reviewer, *AIAA SciTech*, 2025, 2026
- Reviewer, *AIAA Journal of Guidance, Control, and Dynamics*, 2025
- Reviewer, *The Journal of Supercomputing*, 2025

Mentoring & Professional Service

- **Research Mentor (External)** — University of Colorado Boulder
  Mentoring a Master's student on active debris maneuver and on-orbit servicing research — Apr. 2026 – Present
- **Undergraduate Research Mentor** — Georgia Tech
  Guided the implementation of orbital routing problems and paper replication for space logistics — Spring 2026 – Present
- **Invited Speaker** — ADD
  Delivered an invited talk for the 6th Research Officer for National Defense — Sep. 2022
- **Selected Mentor** — ADD
  Mentoring for the 4th Research Officer for National Defense — Oct. 2020 – Jan. 2021

Projects

- Innovative Decision-Making Frameworks for Next-Generation Space Systems — Mitsubishi Electric Corporation, Georgia Tech
  Oct. 2025 – Aug. 2027
- Orbital Staging of Assets (Phase II) — Air Force STTR (Subaward from ThinkOrbital), Georgia Tech
  Oct. 2025 – July 2026
- ATLAS: Advancing Technologies for Logistics Architectures in Space — AFRL/AFOSR, Georgia Tech
  Aug. 2024 – Sep. 2026
- Space Logistics Model for Temporary Deployment of Earth Imaging Satellites — ThinkOrbital, Georgia Tech
  Feb. 2025 – Apr. 2025
- Mission Planning for Swarm of Tiny Satellites — LIG Nex1, KAIST
  Oct. 2023 – Jul. 2024
- Tactical Swarm Unmanned Aerial Vehicle Mission Planning and Autonomous Mission Replanning Technology — LIG Nex1, KAIST
  Oct. 2022 – Jul. 2024
- Modular Architecture Engineering: Extension Considering Practicality — Hyundai, KAIST
  Sep. 2022 – Jul. 2024
- Routing Problems Considering Value System for Exploration Missions — KAIST
  Sep. 2022 – Jul. 2024
- Low Observable Wingman UAV System — ADD
  Nov. 2021 – May. 2022
- Real-Time Decentralized Task Allocation for Multiple and Heterogeneous Unmanned Aerial Systems — AFRL, ADD
  Aug. 2021 – May. 2022
- Autonomous Navigation and Mission Management Technology — ADD
  Dec. 2019 – Aug. 2020